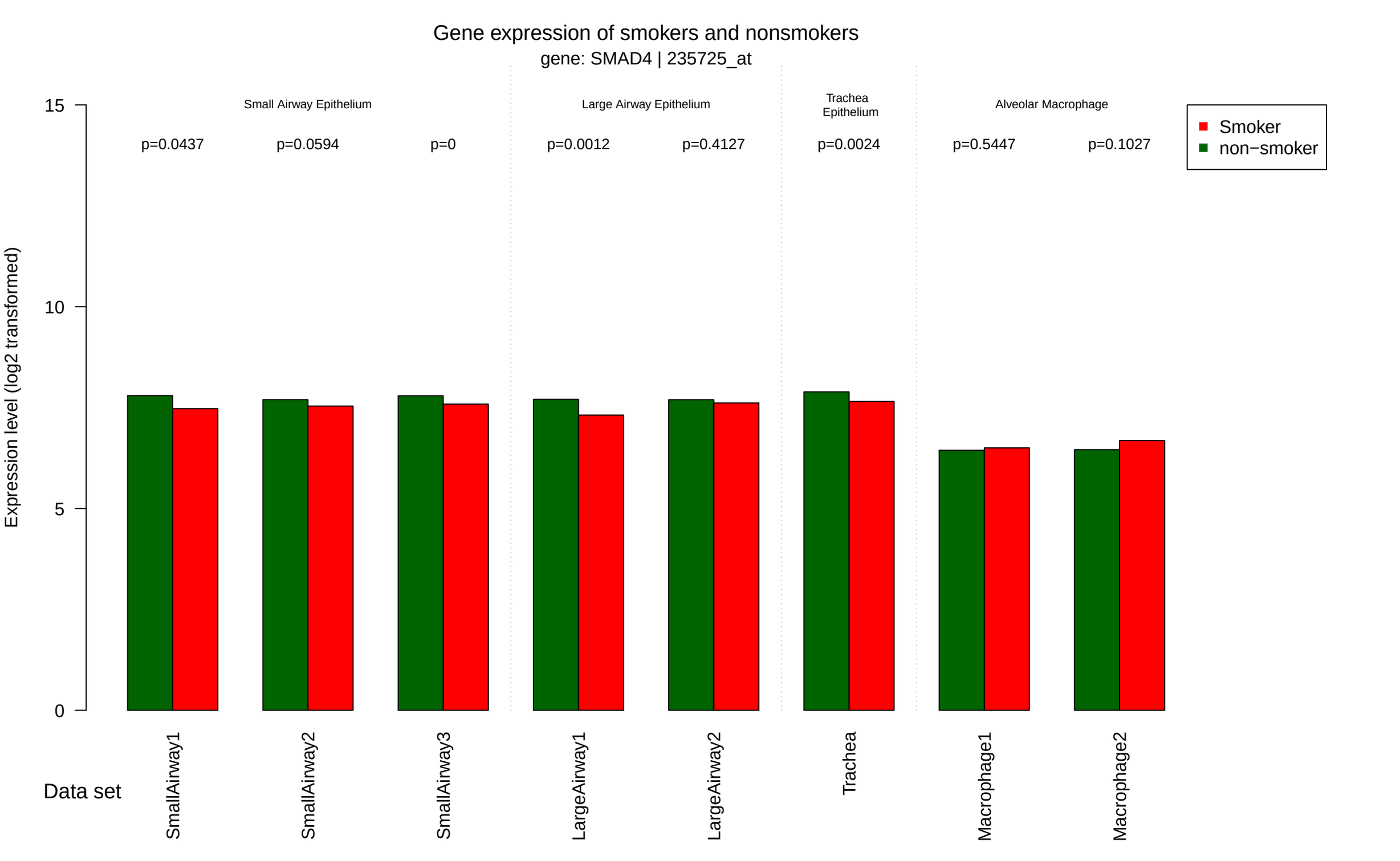
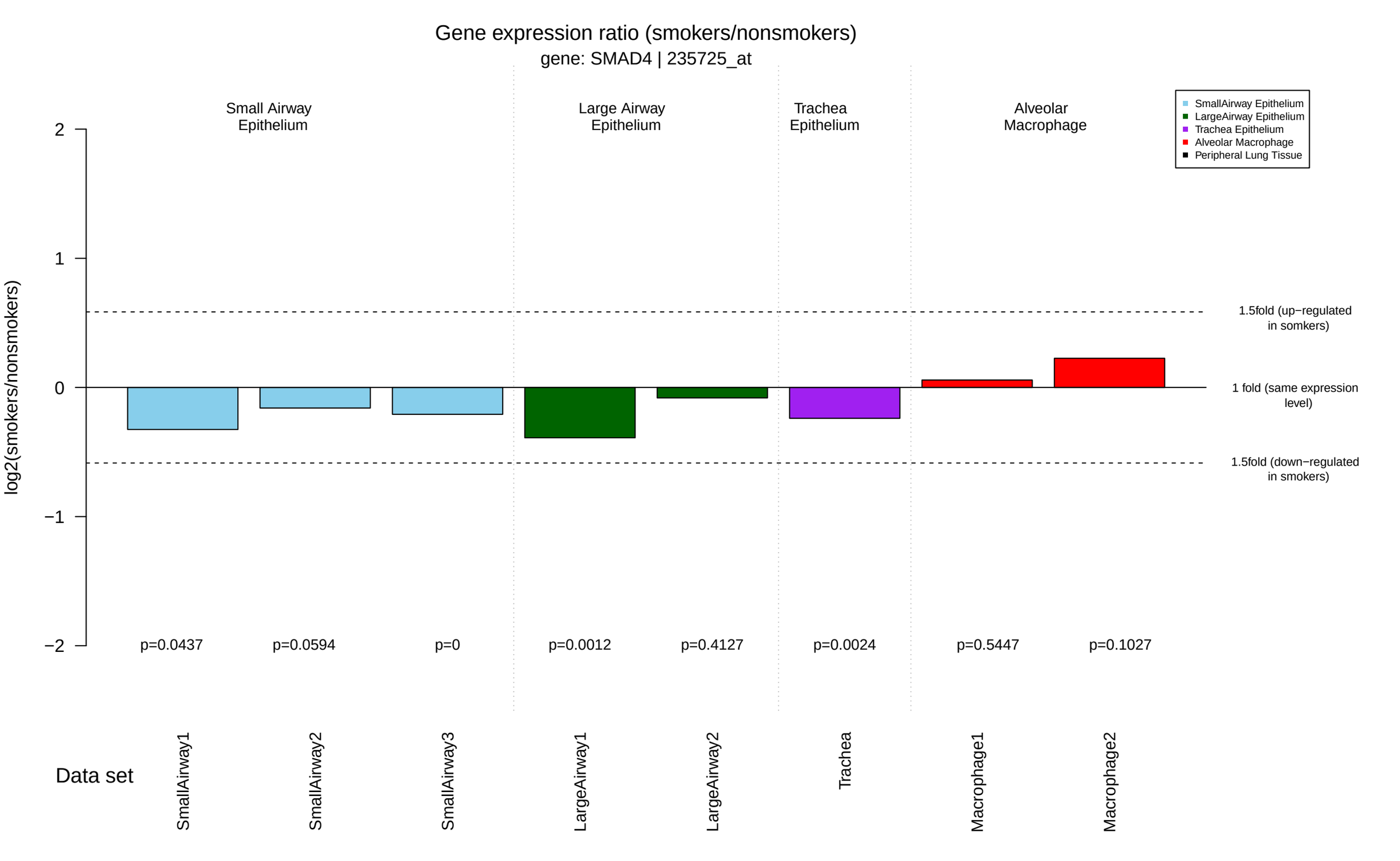
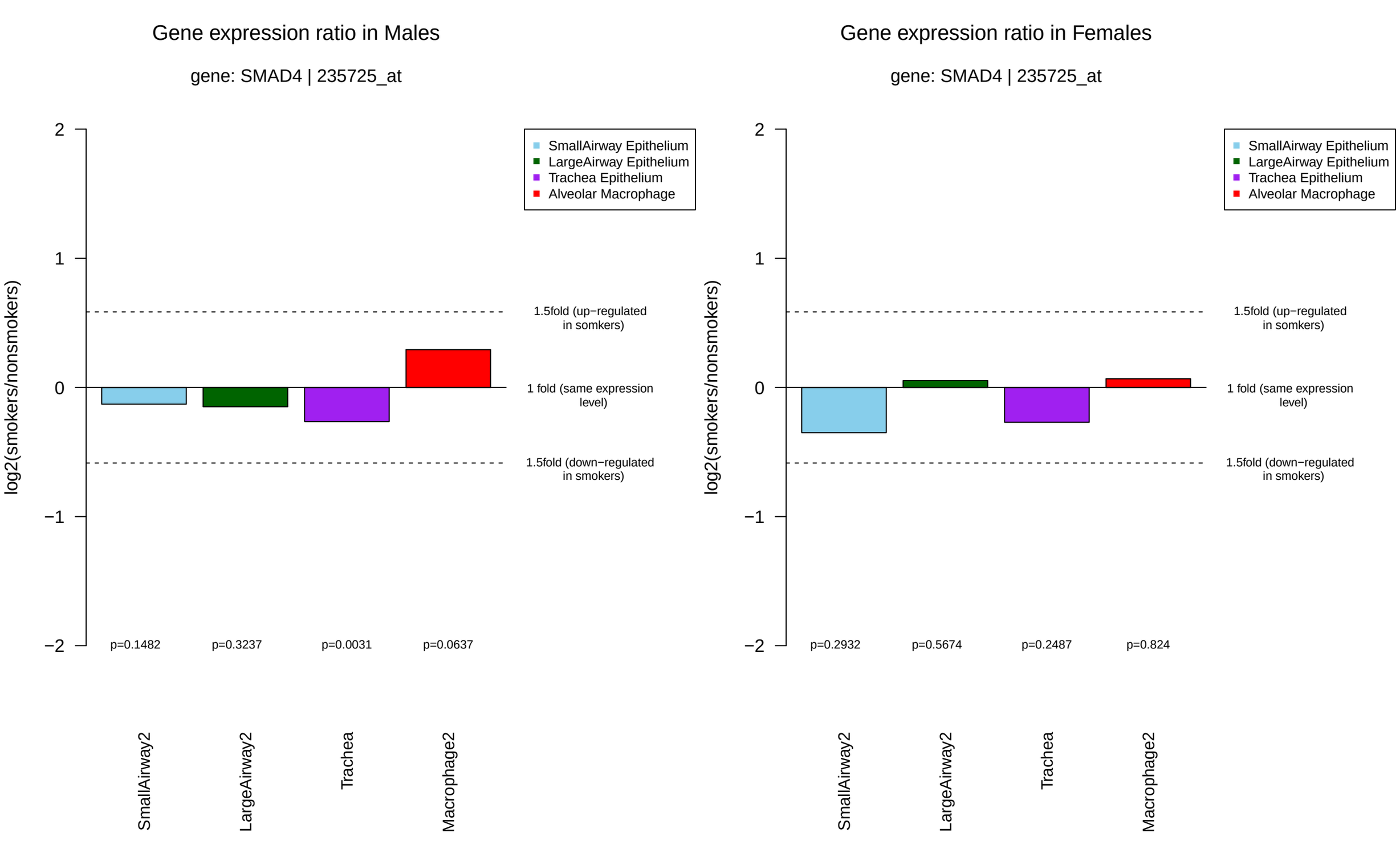
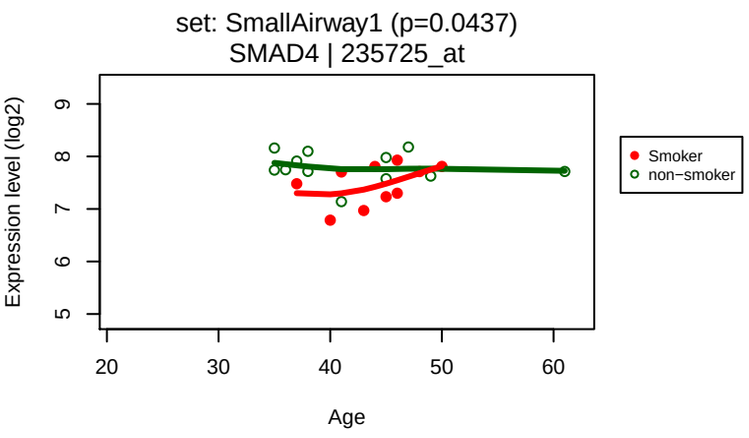
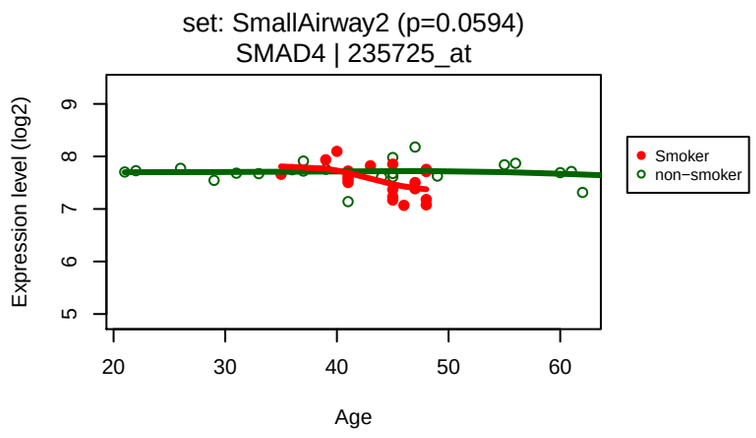
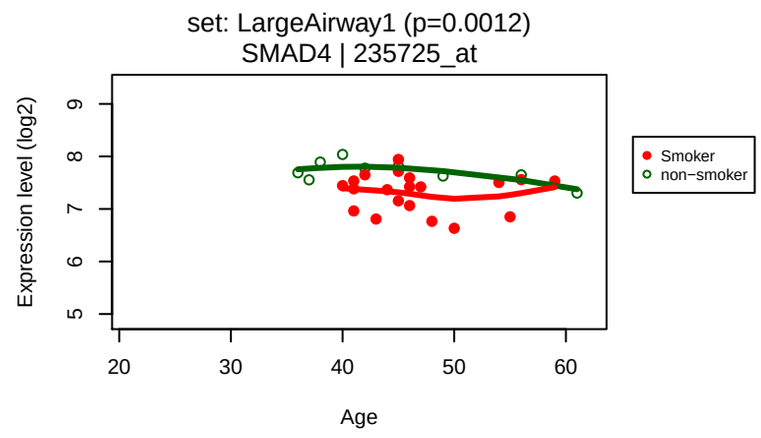

Gene expression of smokers and nonsmokers
gene: SMAD4 | 235725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0437
p=0.0594
p=0
p=0.0012
p=0.4127
p=0.0024
p=0.5447
p=0.1027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

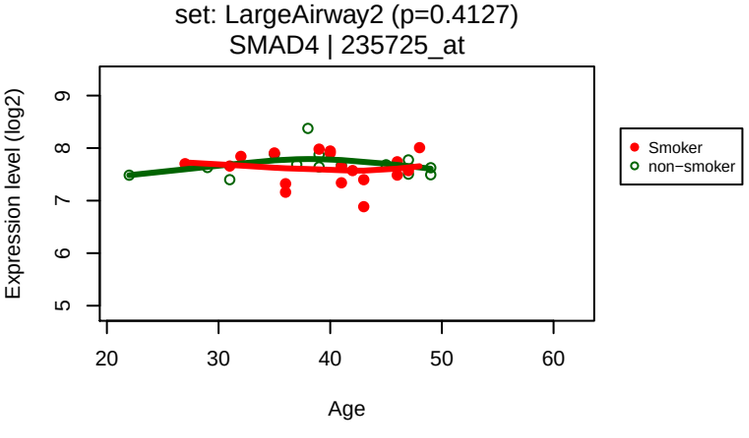

Gene expression ratio (smokers/nonsmokers)
gene: SMAD4 | 235725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0437
p=0.0594
p=0
p=0.0012
p=0.4127
p=0.0024
p=0.5447
p=0.1027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

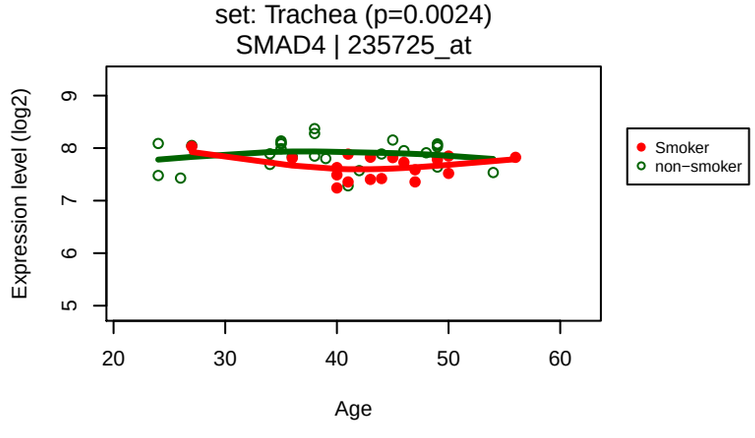

Gene expression ratio in Males
gene: SMAD4 | 235725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1482
p=0.3237
p=0.0031
p=0.0637
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD4 | 235725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2932
p=0.5674
p=0.2487
p=0.824
SmallAirway2
LargeAirway2
Trachea
Macrophage2

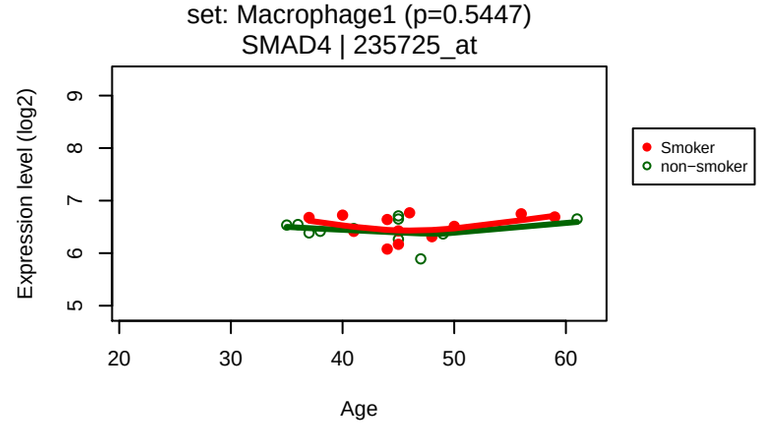

set: SmallAirway1 (p=0.0437)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker

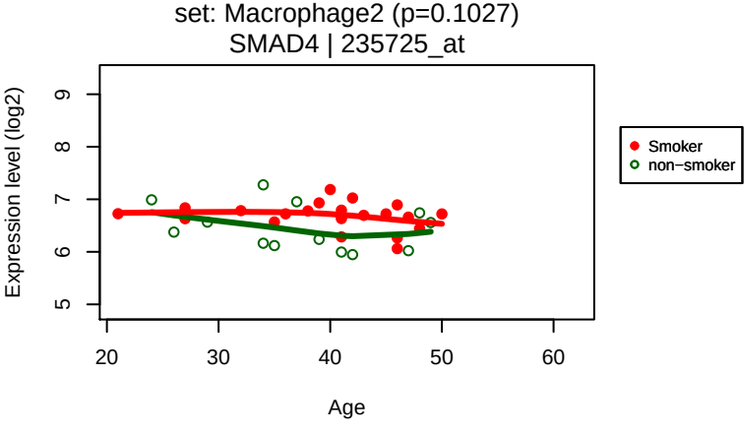

set: SmallAirway2 (p=0.0594)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker

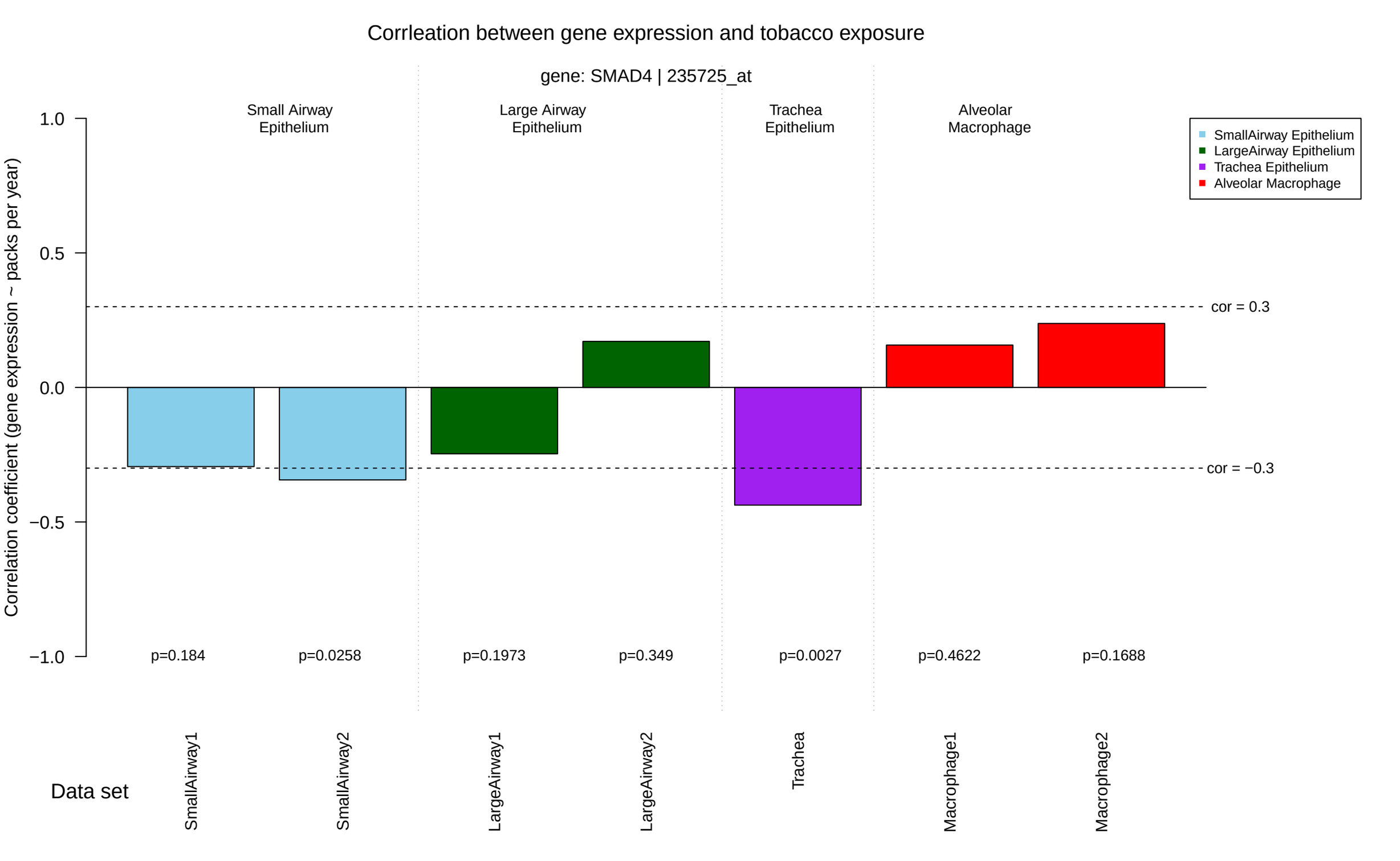

set: LargeAirway1 (p=0.0012)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4127)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0024)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5447)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1027)
SMAD4 | 235725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD4 | 235725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.184
p=0.0258
p=0.1973
p=0.349
p=0.0027
p=0.4622
p=0.1688
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2